

LAB_9

1. Perform the following tasks:

a. Create Student table with following attributes (STUDENT_ID , FIRST_NAME, LAST_NAME, PHONE_NUMBER, MARKS, COURSE_ID).

QUERY:-

```
mysql> create table student(student_id int primary key,first_name varchar(50),last_name varchar(50),phone_number varchar(15),marks decimal(5,2),course_id int );
```

OUTPUT:- Query OK, 0 rows affected (0.04 sec)

b. Create Course table with following attributes (COURSE_ID, COURSE_NAME).

QUERY:-

```
mysql> create table course(course_id int primary key,course_name varchar(50));
```

OUTPUT:- Query OK, 0 rows affected (0.03 sec)

c. Write a SQL statement to insert 8 records with your own value into the tables.

QUERY:-

```
mysql> insert into course(course_id,course_name)values(1,'computer'),(2,'science'),(3,'physic'),(4,'maths'),(5,'chemistry'),(6,'hindi'),(7,'english'),(8,'biology');
```

OUTPUT:- Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

```
mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(1,'ankit','singh','9876543210',90.87,1);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(2,'anup','singh','9876549999',70.87,2);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(3,'pardeep','narwal','9876548899',80.07,3);
Query OK, 1 row affected (0.00 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(4,'sonu','narwal','9876547799',65.07,4);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(5,'pawan','sherwat','9876547799',55.07,5);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(6,'deepak','hooda','9876547700',87.07,6);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(7,'amit','hooda','9878547700',67.07,7);
Query OK, 1 row affected (0.01 sec)

mysql> insert into student(student_id,first_name,last_name,phone_number,marks,course_id)values(8,'rahul','choudhary','9878533700',98.07,8);
Query OK, 1 row affected (0.01 sec)
```

Query:-

Output:-

```
mysql> select*from student;
```

student_id	first_name	last_name	phone_number	marks	course_id
1	ankit	singh	9876543210	90.87	1
2	anup	singh	9876549999	70.87	2
3	pardeep	narwal	9876548899	80.07	3
4	sonu	narwal	9876547799	65.07	4
5	pawan	sherwat	9876547799	55.07	5
6	deepak	hooda	9876547700	87.07	6
7	amit	hooda	9878547700	67.07	7
8	rahul	choudhary	9878533700	98.07	8

8 rows in set (0.00 sec)

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d. Write a query to get the number of students with the same course.

QUERY:- `mysql> select course_id,count(student_id)from student group by course_id;`

course_id	count(student_id)
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1

OUTPUT:- 8 rows in set (0.01 sec)

f. Write a query to get the student name, course name and marks of the students.

QUERY:-

`mysql> select s.first_name as student_name,c.course_name, s.marks from student s join course c on s.course_id=c.course_id;`

OUTPUT:-

student_name	course_name	marks
ankit	computer	90.87
anup	science	70.87
pardeep	physic	80.07
sonu	maths	65.07
pawan	chemistry	55.07
deepak	hindi	87.07
amit	english	67.07
rahul	biology	98.07

8 rows in set (0.00 sec)

g. Write a query to get the Average marks of students course wise.

QUERY:-

`mysql> select c.course_name,avg(s.marks) as avaragemarks from student s join course c on s.course_id=c.course_id group by c.course_name;`

OUTPUT:-

course_name	avaragemarks
computer	72.970000
hindi	78.970000
physic	80.070000
maths	81.570000
english	67.070000

5 rows in set (0.00 sec)

2. Create database for hospital management system & Perform the following tasks:

a. Create HEALTH CARE WORKERS table with following attributes (EMPLOYEE_ID , FIRST_NAME, LAST_NAME,EMAIL, PHONE_NUMBER, HIRE_DATE, SALARY, DESIGNATION).

QUERY:-

```
mysql> create table healthcare_worker(employee_id int primary key,first_name varchar(50),last_name varchar(50),email varchar(50),phone_number varchar(15),hire_date date,salary int,designation varchar(50));
```

OUTPUT:-

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
email	varchar(50)	YES		NULL	
phone_number	varchar(15)	YES		NULL	
hire_date	varchar(50)	YES		NULL	
salary	int	YES		NULL	
designation	varchar(50)	YES		NULL	

8 rows in set (0.00 sec)

b. Create PATIENT table with following attributes (PATIENT_ID,NAME, PHONE_NUMBER).

QUERY:-

```
mysql> create table patient(patient_id int primary key,name varchar(50),phone_number varchar(50));
```

OUTPUT:-

Field	Type	Null	Key	Default	Extra
patient_id	int	NO	PRI	NULL	
name	varchar(50)	YES		NULL	
phone_number	varchar(15)	YES		NULL	

3 rows in set (0.00 sec)

c. Write a SQL statement to insert 10 records with your own value into the tables.

QUERY:-

OUTPUT:-

employee_id	first_name	last_name	email	phone_number	hire_date	salary	designation
1	ram	pal	pal@gmail.com	9876543210	12/3/24	20000	surgen
2	summo	bhoir	bhoir@gmail.com	9876542222	20/3/24	15000	trainee
3	namira	mulla	mulla@gmail.com	9876543333	20/3/24	15000	trainee
4	sidhi	jain	jain@gmail.com	9876544444	20/3/24	15000	trainee
5	helsiki	heli	helsiki@gmail.com	9876523444	20/4/24	18000	jr.doctor
6	raja	kallu	kallu@gmail.com	9876523884	27/4/24	22000	sr.doctor
7	kajal	prasad	prasad@gmail.com	9876333884	27/4/24	22000	sr.doctor
8	mansi	singh	singh@gmail.com	9899333899	1/4/24	10000	nuars
9	vansh	singh	singh@gmail.com	9899333884	23/4/24	26000	heart_surgon
10	raj	sinha	sinha@gmail.com	9899333877	1/4/24	10000	wardboy

10 rows in set (0.00 sec)

patient_id	name	phone_number
1	ankit	9876543210
2	siya	987667890
3	geeta	9876612345
4	prem	9876090909
5	manju	9876098887
6	golu	1212098887
7	kallu	1276768887
8	ritika	1276764545
9	riya	1275656545
10	jacky	1275656500

10 rows in set (0.00 sec)

d. Write a query to get the names (first_name, last_name), Designation, salary.

QUERY:-

```
mysql> select first_name, last_name, Designation, salary from healthcare_worker;
```

OUTPUT:-

first_name	last_name	Designation	salary
ram	pal	surgen	20000
summo	bhoir	trainee	15000
namira	mulla	trainee	15000
sidhi	jain	trainee	15000
helsiki	heli	jr.doctor	18000
raja	kallu	sr.doctor	22000
kajal	prasad	sr.doctor	22000
mansi	singh	nuars	10000
vansh	singh	heart_surgon	26000
raj	sinha	wardboy	10000

10 rows in set (0.00 sec)

e. Write a query to get the number of employees with the same Designation

QUERY:-

```
mysql> SELECT DESIGNATION,COUNT(EMPLOYEE_ID) FROM HEALTHCARE_WORKER GROUP BY DESIGNATION;
```

OUTPUT:-

DESIGNATION	COUNT(EMPLOYEE_ID)
surgen	1
trainee	3
jr.doctor	1
sr.doctor	2
nuars	1
heart_surgon	1
wardboy	1

7 rows in set (0.00 sec)

f. Write a query to get employee name who are getting salary more than 25000.

QUERY:-

```
mysql> SELECT FIRST_NAME AS EMPLOYEE_NAME FROM HEALTHCARE_WORKER WHERE SALARY>25000;
```

OUTPUT:-

```

+-----+
| EMPLOYEE_NAME |
+-----+
| vansh         |
+-----+
1 row in set (0.00 sec)

```

g. Fetch HEALTH CARE WORKERS name using their employee id.

QUERY:-

```
mysql> select first_name as NAME from healthcare_worker where employee_id in(1,3,6,8,4);
```

OUTPUT:-

```

+-----+
| NAME |
+-----+
| ram   |
| namira |
| sidhi |
| raja  |
| mansi |
+-----+
5 rows in set (0.00 sec)

```

3.Consider two tables, customers and orders, with the following structures:

Customers Table: customer_id (Primary Key) first_name Last_name

Orders Table: order_id (Primary Key) customer_id (Foreign Key) order_date Total_amount

Write an SQL query to retrieve the first and last names of customers along with the order date and total amount of their orders.

Use an INNER JOIN to connect the two tables.

QUERY:-SELECT

CUSTOMERS.FIRST_NAME,CUSTOMERS.LAST_NAME,ORDERS.ORDER_DATE,ORDERS.TOTAL_AMOUNT

FROM CUSTOMER JOIN

ORDER ON CUSTOMERS.CUSTOMER_ID=ORDERS.CUSTOMER_ID;

OUTPUT:-

first_name	last_name	order_date	total_amount
Rajesh	Sharma	2024-09-01	5000.00
Priya	Kumar	2024-09-02	7500.50
Anjali	Verma	2024-09-03	12000.75
Sunil	Patil	2024-09-04	3000.00
Meena	Gupta	2024-09-05	4500.25

5 rows in set (0.00 sec)

4.Consider two tables, departments and employees, with the following structures:

Departments Table: department_id (Primary Key) department_name

Employees Table: employee_id (Primary Key) first_name last_name
department_id (Foreign Key)

Write an SQL query to retrieve a list of all departments and the names of employees who belong to each department. Use a LEFT JOIN to include departments that have no employees.

QUERY:-

SELECT

DEPARTMENTS.DEPARTMENT_NAME, EMPLOYEES.FIRST_NAME,
EMPLOYEES.LAST_NAME

FROM DEPARTMENTS LEFT JOIN

EMPLOYEES ON DEPARTMENTS.DEPARTMENT_ID = EMPLOYEES.DEPARTMENT_ID;

OUTPUT:-

DEPARTMENT_NAME	FIRST_NAME	LAST_NAME
HR	Sunita	Rao
Engineering	Rakesh	Mehta
Marketing	Priya	Deshmukh
Finance	NULL	NULL

4 rows in set (0.00 sec)

4. Write a program to show JDBC connection with MYSQL and perform the following operations:

Create table Customer with following fields:

Custno, Custame,Custaddress,Phoneno, City, Pincode, Country

Insert 5 records in Customer table.

- Insert values
- Delete values
- update city name Shimla to Shilong.
- Show table in the console

PROGRAM

//CONNECTJDBC

package Jdbc;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;

public class ConnectJDBC {

```

    public static Connection getConnection() throws SQLException {
        try
        {
            Class.forName("com.mysql.cj.jdbc.Driver");

            return
            DriverManager.getConnection("jdbc:mysql://localhost:3306/anudip", "root", "vansh@2304");
        } catch (ClassNotFoundException e)
        {
            e.printStackTrace();

            throw new SQLException("MySQL JDBC Driver not found.");
        }
    }
}

```

//JDBCOPERATION1

```

package Jdbc;

import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class JdbcOperation1 {

```

```

    public class DBOperations {

        public static void main(String[] args) {

            Connection conn = null;

```

```
Statement stmt = null;
```

```
ResultSet rs = null;
```

```
try {
```

```
    // Establish a connection to the database
```

```
    conn = ConnectJDBC.getConnection();
```

```
    stmt = conn.createStatement();
```

```
    // Create the Customer table if it doesn't exist
```

```
    String createTableSQL = "CREATE TABLE IF NOT EXISTS Customer ("
```

```
+
```

```
        "Custno INT AUTO_INCREMENT PRIMARY KEY, " +
```

```
        "Custname VARCHAR(100), " +
```

```
        "Custaddress VARCHAR(255), " +
```

```
        "Phoneno VARCHAR(15), " +
```

```
        "City VARCHAR(50), " +
```

```
        "Pincode VARCHAR(10), " +
```

```
        "Country VARCHAR(50))");
```

```
    stmt.execute(createTableSQL); // Execute table creation
```

```
    System.out.println("Customer table created.");
```

```
    // Insert three records into the Customer table
```

```
    String insertSQL1 = "INSERT INTO Customer (Custname,  
Custaddress, Phoneno, City, Pincode, Country) " +
```

```
        "VALUES ('ANKIT SINGH', 'AZAD NAGAR ',  
'9876543210', 'BOISAR', '452001', 'India')";
```

```
    stmt.executeUpdate(insertSQL1); // Execute the insert operation
```

```
String insertSQL2 = "INSERT INTO Customer (Custname,
Custaddress, Phoneno, City, Pincode, Country) " +
                    "VALUES ('RITIK SINGH', 'CHINA TOWN',
'9123456789', 'Bengaluru', '560001', 'India')";
stmt.executeUpdate(insertSQL2); // Execute the insert operation
```

```
String insertSQL3 = "INSERT INTO Customer (Custname,
Custaddress, Phoneno, City, Pincode, Country) " +
                    "VALUES ('SAURABH SINGH', ' BHEEM NAGAR',
'9988776655', 'VARANASI', '600002', 'India')";
stmt.executeUpdate(insertSQL3); // Execute the insert operation
```

```
System.out.println("3 records inserted into Customer table.");
```

```
// Retrieve and display all records from the Customer table
```

```
String selectSQL = "SELECT * FROM Customer";
```

```
rs = stmt.executeQuery(selectSQL);
```

```
while (rs.next()) {
```

```
    int custno = rs.getInt("Custno");
```

```
    String custname = rs.getString("Custname");
```

```
    String custaddress = rs.getString("Custaddress");
```

```
    String phoneno = rs.getString("Phoneno");
```

```
    String city = rs.getString("City");
```

```
    String pincode = rs.getString("Pincode");
```

```
    String country = rs.getString("Country");
```

```
    System.out.println("Custno: " + custno + ", Custname: " +
custname + ", Custaddress: " + custaddress +
```

```
        ", Phoneno: " + phoneno + ", City: " + city +  
        ", Pincode: " + pincode + ", Country: " + country);  
  
    }  
  
    } catch (SQLException e) {  
        e.printStackTrace();  
    } finally {  
        // Close resources  
        try {  
            if (rs != null) rs.close();  
            if (stmt != null) stmt.close();  
            if (conn != null) conn.close();  
        } catch (SQLException e) {  
            e.printStackTrace();  
        }  
    }  
}  
  
}
```